

UNIVERSITAS GADJAH MADA
FAKULTAS TEKNIK
DEPARTEMEN TEKNIK NUKLIR DAN TEKNIK FISIKA

**DOCUMENT
DESCRIPTION OF FINAL PROJECT**

A. Official Legalization

Name	:	Arcelia Maleeka Haifa Maajid
Student ID	:	22/504603/TK/55201
Topic	:	Radiotherapy Physics
Title (Indonesia)	:	Pengaruh Variasi Energi Proton terhadap RBE pada <i>Entrance Region Proton Therapy</i> dengan Model 3D Glioblastoma (U87)
Title (English)	:	<i>Investigation of Energy-Dependent Variations in Proton RBE at the Entrance Region in 3D U87 Glioblastoma</i>
Field (Discipline)	:	Radiotherapy Physics
Keyword (Indonesia)	:	Terapi Proton, Radiobiologi, Glioblastoma, Kultur Sel 3D, Relative Biological Effectiveness (RBE), Energi Proton, Entrance Region
Keyword (English)	:	<i>Proton Therapy, Radiobiology, Glioblastoma, 3D Cell Culture, Relative Biological Effectiveness (RBE), Proton Energy, Entrance Region</i>
Proposed Advisors		
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Start Date	:	01/07/2026
End Date	:	01/12/2026

Approved by

Head of the Nuclear Engineering Study Program



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B. Description of Final Project

1. Introduction

Glioblastoma is a highly aggressive tumor that is difficult to treat due to its resistance to radiation [1]. As a result, proton therapy is increasingly used because it delivers a concentrated dose to the tumor while minimizing damage to the healthy tissue [2]. To link physical dose with biological response, the relative biological effectiveness (RBE) is used, where a constant RBE value of 1.1 is typically applied in clinical practice [3]. Cell culture is an in vitro method for growing and studying cells outside an organism under controlled environmental conditions. 3D cell culture was developed to address the shortcomings of 2D cell culture, allowing cells to form aggregates or structures that more closely resemble the organization of tissues in vivo [4]. Recent studies have developed spatially resolved 3D cell cultures that more accurately represent the biological response to protons and have found a higher RBE value of 1.15 for the glioblastoma cell line (U87) at the entrance region for a single proton energy of 100 MeV [5]. This research aims to evaluate how variations in proton energy affect RBE values in 3D cell cultures when irradiated at the same position in the entrance region.

2. Research Questions

The problem formulation in this research includes:

1. What is the proton relative biological effectiveness (RBE) in the entrance region as measured experimentally using 3D cell culture, compared with the clinically assumed value of 1.1?
2. How do variations in proton energy at identical irradiation positions within the entrance region influence the measured proton relative biological effectiveness (RBE) in 3D cell cultures?

3. Objectives

The objectives of this research include:

1. To evaluate proton RBE values in 3D cell cultures in the entry region compared to the clinical assumption at 1.1.
2. To assess the changes in RBE values due to variation in proton energy at the same irradiation position in the entry region.

4. Fundamental Theory

Glioblastoma is a malignant brain tumor with high radioresistance. This resistance is attributable to its biological structure, originating from various cell types, including neural stem cells, that differentiate into broad cell forms (pleomorphic), commonly known as multiform [1]. Protons are an effective modality because their energy deposition characteristics follow the path length of the Bragg curve [2]. To evaluate the cellular response to radiation, in vitro cell culture models are commonly used. Three-dimensional cell culture models represent the tumor microenvironment, which includes interactions between cells and the extracellular matrix, as well as an oxygen gradient and nutrient gradients [4]. In recent studies, three-dimensional cell cultures have shown a radiation response characterized by a tendency toward radioresistance and greater sensitivity to protons than to x-rays [5]. Since protons have a relatively low linear energy transfer (LET) and high kinetic energy in the entrance region, the radiation quality can be simplified by assuming a constant RBE of 1.1 [3]. However, energy variations still affect the ionization distribution at the microscopic scale, which contributes to clustered DNA



damage. These ionization patterns induce complex double-strand breaks that are difficult to repair, leading to detrimental effects such as apoptosis [3], [6].

5. Work Plan

This research begins by modeling proton simulations with the TOPAS software to determine the approximate depth of the entrance region in the Bragg curve to match the setup of the irradiation conditions used [7]. Afterwards, following the reference study, U87 cells will be cultured in a monolayer (2D) and a Matrigel-based 3D bio-phantom to model tumor microenvironment conditions [5]. Irradiation will be accomplished using monoenergetic proton beams of 120, 150, and 200 MeV by adjusting the range for targets in the head or brain by placing a 1–2 cm thick PMMA layer according to the target entry region [5], [8]. The biological response of cells to proton radiation will be assessed using a clonogenic assay to measure their ability to form colonies. Then, the survival fraction value will be modeled using a Linear-Quadratic model, while the RBE10% will be determined with a cell survival rate limit of 10% [3], [5], [9], [10].

6. References

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Lampiran: Histori alur persetujuan

No	Jabatan	Nama	Jenis	Tanggal Disetujui
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